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Mean surface density flux constrains the amplitude of AMOC
decadal variabilityZhenbang Xu^{1,2} , Shujun Li^{1,3,*} , Chuan-Yang Wang^{1,4} and Lixin Wu^{1,3}¹ Key Laboratory of Physical Oceanography/Frontier Science Centre for Deep Ocean Multispheres and Earth system, Ocean University of China, Qingdao, People's Republic of China² Academy of Future Ocean, Ocean University of China, Qingdao, People's Republic of China³ Laoshan Laboratory, Qingdao, People's Republic of China⁴ College of Oceanic and Atmospheric Sciences, Ocean University of China, Qingdao, People's Republic of China

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E-mail: lishujun@ouc.edu.cn**Keywords:** Atlantic meridional overturning circulation, decadal variability, surface density flux, mean–variability coupling, buoyancy forcingSupplementary material for this article is available [online](#)**Abstract**

Decadal variability of the Atlantic meridional overturning circulation (AMOC-DV) is a major source of climate variability in the Atlantic and exerts broad influence on the climate system. However, the factors controlling its amplitude across climate models and background climate states remain insufficiently understood. Here, we show that AMOC-DV amplitude is closely constrained by the mean surface density flux over the subpolar North Atlantic across CMIP6 models and climate states, including preindustrial, historical, and future warming climates. This emergent relationship is further supported by the CESM2 Large Ensemble. Although internal variability can produce spread in AMOC-DV amplitude among ensemble members under similar mean density flux conditions, but is insufficient to explain the systematic changes in AMOC-DV amplitude across climate states, which remain closely associated with changes in mean density flux. This constraint mainly arises through two synergistic pathways. The mean buoyancy forcing sets the dynamical gain of the overturning circulation, controlling the efficiency with which density flux anomalies drive AMOC variability, while also modulating the variance of density flux fluctuations themselves. Together, these processes link the mean state to AMOC variability, with stronger mean surface density flux corresponding to a higher-gain and more strongly forced regime that supports larger AMOC-DV amplitude. Our findings reveal a fundamental control of subpolar buoyancy forcing on AMOC-DV, implying that accurate simulation of subpolar buoyancy forcing is critical for reliable projections of AMOC-related climate variability under climate change.

1. Introduction

Atlantic meridional overturning circulation (AMOC) is a fundamental component of the climate system, which transports heat meridionally across hemispheres and regulates the freshwater distribution, biogeochemical processes and marine ecosystems (Buckley and Marshall 2016, Brown *et al* 2021, Jackson *et al* 2022). Beyond its projected weakening under global warming (Liu *et al* 2017, Liu *et al* 2020), the AMOC exhibits pronounced decadal variability (AMOC-DV), which influences North Atlantic sea surface temperature (SST), atmospheric circulation,

and hydroclimate (Zhang *et al* 2019, Megann *et al* 2021). For example, a strengthened AMOC enhances northward heat transport, leading to warmer SST in the North Atlantic and potentially reinforcing the positive phase of AMV (Delworth *et al* 2017, Hand *et al* 2020). This shift may subsequently trigger increased Sahel summer rainfall and weakened vertical wind shear over the main development region of Atlantic hurricane, thereby affecting regional climate (Li *et al* 2025). Clarifying what controls the strength of AMOC-DV is therefore essential for understanding and predicting regional climate variability under future climate change.

AMOC-DV is generally interpreted as a coupled oceanic oscillatory mode, arising from interactions among surface buoyancy forcing, deep-water formation processes, and ocean dynamical adjustment (Delworth *et al* 1993, Dong and Sutton 2005, Gao *et al* 2025). Specifically, surface heat and freshwater fluxes generate density anomalies in deep convection regions, which influence deep-water formation, overturning strength, and associated ocean heat transport (Kostov *et al* 2019, Liu and Fedorov 2022). These circulation anomalies subsequently evolve through basin-scale ocean dynamical adjustment, including baroclinic Rossby wave propagation and advective feedbacks, which regulate the spatial evolution and characteristic timescale of the AMOC-DV (Buckley *et al* 2012, Muir and Fedorov 2017). Together, these studies have highlighted the importance of coupled buoyancy and dynamical processes in shaping the generation and evolution of AMOC-DV, while the factors controlling its amplitude across different climate states remain less well understood.

Previous studies have examined how the AMOC-DV responds to global warming (Cheng *et al* 2016, Ma *et al* 2021). Multi-model analyses suggest that global warming generally weakens the amplitude of AMOC-DV and shortens its dominant period. These changes have been attributed to enhanced upper-ocean stratification under warming conditions, which accelerates baroclinic Rossby wave propagation and basin-scale ocean adjustment. The faster adjustment reduces the delay time of ocean dynamical feedbacks, leading to a shorter oscillation timescale and potentially a weaker AMOC-DV amplitude. These interpretations emphasize the importance of ocean dynamical adjustment in regulating the amplitude and timescale of AMOC-DV under climate change. However, an important inconsistency remains when comparing the responses of AMOC-DV amplitude and period across climate models (Cheng *et al* 2016). Although ensemble-mean analyses indicate a coherent relationship between reduced period and reduced amplitude of AMOC-DV, examination of individual models reveals a more complex picture. The projected weakening of AMOC-DV amplitude is highly consistent among models, while the response of the oscillation period exhibits large inter-model differences, with some models showing pronounced shortening while others exhibiting little change or even lengthening period. This discrepancy suggests that the processes controlling the oscillation timescale may not fully determine the magnitude of AMOC-DV. Although ocean dynamical adjustment governs how AMOC anomalies propagate and evolve, an additional mechanism may be required to explain why the amplitude reduction is much more consistent across climate models.

Inter-model differences in the AMOC-DV amplitude have been linked to differences in the mean overturning state, suggesting a potential connection

between the background circulation and variability strength (Kim *et al* 2023, Ferster *et al* 2025). However, the mean AMOC strength is an emergent property of the coupled climate system rather than a direct energy source sustaining variability. From an energetic perspective, surface buoyancy forcing regulates the availability and redistribution of oceanic available potential energy by modifying the density structure of the upper ocean. Expressed as the surface density flux, this forcing characterizes the background buoyancy conditions that govern deep convection and the strength of the overturning circulation. If AMOC-DV behaves as a buoyancy-mediated oscillatory mode, its amplitude should depend not only on the efficiency of dynamical adjustment processes but also on the magnitude of the background buoyancy energy available to sustain variability. This raises a fundamental question: does the mean surface density flux impose an energetic constraint on the AMOC-DV amplitude across different models and climate states? Addressing this question is important because the AMOC-DV amplitude influences the persistence and detectability of North Atlantic climate signals, as well as the relative roles of internal variability and externally forced changes under climate warming. A systematic weakening of AMOC-DV may therefore affect both decadal climate predictability and the emergence of anthropogenic climate signals in the North Atlantic.

Global warming experiments provide a useful framework to test this hypothesis, as they alter surface heat and freshwater fluxes, and hence the mean buoyancy forcing over deep convection regions. Here, using multi-model and multi-scenario simulations from the coupled model intercomparison project phase 6 (CMIP6; Eyring *et al* 2016) spanning preindustrial to late twenty-third-century climates, together with Community Earth System Model version 2 Large Ensemble (CESM2-LE) simulations, we show that the AMOC-DV amplitude is closely constrained by the mean buoyancy forcing over the subpolar North Atlantic across models and climate states. We further identify the dynamical mechanisms underlying this constraint.

2. Data and methods

2.1. Data

2.1.1. CMIP6 outputs

We use simulations from the CMIP6 archive to investigate the AMOC-DV under different climate forcing scenarios. We include all available CMIP6 models that provide the required monthly outputs continuously over the period 2015–2299 under future warming scenarios. Seven models meet these criteria, including six models with one available ensemble member (r1i1p1f1) and one model with two additional ensemble members (r1i1p1f2 and r1i1p3f1). In total, eight ensemble members are analyzed (table S1). All

ensemble members are retained in the analyses and given equal weight, with the additional members from the same model providing an opportunity to assess the extent to which internal variability may influence the results.

Several CMIP6 experiments are analyzed to characterize AMOC-DV under different forcing conditions. The piControl simulations, which maintain atmospheric CO₂ concentrations at preindustrial levels, are used to quantify the unforced internal variability of the climate system. To reduce the influence of model drift, only the final 200 years of each piControl simulation are analyzed. Historical simulations, which include both anthropogenic and natural forcings, are analyzed for the period 1850–2014. Future climate evolution is assessed using two Shared Socioeconomic Pathway (SSP) scenarios, SSP1-2.6 and SSP5-8.5 (hereafter SSP126 and SSP585), covering 2015–2299. These scenarios represent low- and high-emission pathways, with radiative forcing reaching approximately 2.6 and 8.5 W m⁻² by 2100, respectively.

The analysis is based on a suite of ocean circulation, hydrographic, and surface flux variables. The AMOC strength is derived from the meridional overturning streamfunction (msftmz/msfty), with meridional ocean velocity (vo) used for additional circulation diagnostics where applicable. Surface density fluxes are calculated from surface components, including latent heat flux (hfls), sensible heat flux (hfss), upward and downward longwave radiation fluxes (rlus, rlds), upward and downward shortwave radiation fluxes (rsus, rsds), precipitation (pr), evaporation (evs), freshwater into sea from river (friver), sea-ice-related freshwater flux (fsitherm), sea surface temperature (tos) and sea surface salinity (sos). Potential density is calculated from seawater salinity (so) and potential temperature (thetao), and the ocean mixed layer depth (MLD) is obtained from mlotst. All model outputs are remapped onto common grids to facilitate inter-model comparison. Surface and horizontally distributed variables are interpolated onto a common 1° × 1° horizontal grid using bilinear interpolation. Three-dimensional ocean fields are first interpolated horizontally using bilinear interpolation and then vertically using linear interpolation onto a grid with 1° × 1° horizontal resolution and 100-m vertical spacing. Meridional overturning streamfunctions are linearly interpolated onto a common latitude–depth grid with 1° latitudinal resolution and 100 m vertical spacing.

2.1.2. CESM2 large ensemble

We use the CESM2-LE to further investigate the relationship between surface density flux and AMOC-DV and to assess the potential influence of internal variability on this relationship. CESM2-LE provides a large ensemble of fully coupled climate simulations with identical external forcing but perturbed

initial conditions, thereby allowing the role of internal climate variability to be examined across ensemble members (Danabasoglu *et al* 2020). We analyze a 50-member subset of CESM2-LE, including the available simulations following the CMIP6 Historical and SSP370 forcing protocols. The Historical simulations cover 1850–2014, while the SSP370 simulations cover 2015–2100. The AMOC strength is derived from the meridional overturning circulation streamfunction (MOC). Surface density fluxes are calculated from surface components, including evaporation (EVAP_F), surface heat flux (SHF), precipitation (PRECT), river runoff (ROFF_F), and sea-ice-related freshwater flux (siflwbob), together with SST and sea surface salinity (SSS). All CESM2-LE outputs are remapped using the same interpolation procedures applied to the CMIP6 datasets.

2.2. Methods

2.2.1. Calculation of AMOC-DV

The AMOC streamfunction represents the meridional overturning circulation as a function of latitude and depth, obtained by vertically integrating meridional transport across the Atlantic basin (figure S1). The AMOC strength is defined as the maximum value of ocean meridional overturning streamfunction below 500 m between 40°–50° N. The meridional overturning mass streamfunction in depth coordinates is calculated as:

$$\psi(z) = \int_{x_w}^{x_e} \int_{-H}^z \rho v(x, z) dz dx \quad (1)$$

where v represents the meridional velocity, ρ is the potential density, z is a vertical coordinate that increases upward, H denotes the water column height from the seafloor to the surface, and x_w and x_e denote the westward and eastward boundaries of the Atlantic basin at a given depth, respectively.

For CanESM5, the directly provided streamfunction output is only available until 2180. Therefore, the AMOC index after 2180 under SSP585 is reconstructed using equation (1) based on available ocean fields. The reconstructed AMOC-DV agrees well with the original index derived from output during the overlapping period (2015–2180; $r = 0.96$), allowing subsequent analysis throughout the SSP585 simulation period.

To quantify AMOC-DV, the AMOC index derived from the overturning streamfunction is first detrended by removing a quadratic trend and then filtered using a 10–100 year bandpass filter to isolate decadal-to-multidecadal variability. This frequency range is chosen to include the dominant timescales of AMOC variability across different models and forcing scenarios while excluding higher-frequency fluctuations and long-term forced trends. The amplitude of AMOC-DV is quantified as the standard deviation of the filtered AMOC index.

2.2.2. Surface density flux estimate

Surface heat flux Q_{net} represents the net exchange of thermal energy between ocean and atmosphere, comprising radiative and turbulent components:

$$Q_{\text{net}} = Q_{\text{SW}} - Q_{\text{LW}} + Q_{\text{LH}} + Q_{\text{SH}} \quad (2)$$

where Q_{SW} is net shortwave radiation, Q_{LW} is net longwave radiation, Q_{LH} is latent heat flux and Q_{SH} is sensible heat flux.

Surface freshwater flux represents the net freshwater exchange at the ocean-atmosphere interface and is calculated as the sum of precipitation (P), river runoff (R) and ice melt (I) minus evaporation (E):

$$F = P + R + I - E. \quad (3)$$

Surface density flux is calculated as follows (Schmitt et al 1989, Gao et al 2024):

$$F_d = F_t + F_h \quad (4)$$

where F_d is the surface density flux, F_t is the thermal density flux and F_h is the haline density flux. These are calculated as following:

$$F_t = -\alpha \cdot \frac{Q_{\text{net}}}{C_p}, F_h = -\rho(0, T) \cdot \beta \cdot \frac{F \cdot S}{1 - S} \quad (5)$$

where $\rho(0, T)$ is the density of freshwater with temperature of T , S is the salinity, α is the thermal expansion coefficient, estimated as $\alpha = -\frac{1}{\rho} \frac{\partial \rho}{\partial T}$; β is the haline contraction coefficient, estimated as $\beta = \frac{1}{\rho} \frac{\partial \rho}{\partial S}$; C_p is the heat capacity of water, estimated as $c_p = \frac{\partial h}{\partial T}$, h is the specific enthalpy. All fluxes are defined as positive into the ocean.

3. Results

3.1. Systematic shift in AMOC-DV across models and climate states

Across CMIP6 simulations, AMOC-DV generally weakens with progressive warming (figure 1). We first calculate the AMOC-DV index for each model under five climate states, including piControl, concatenated historical and transient SSP126 simulations (2015–2099), concatenated historical and transient SSP585 simulations (2015–2099), extended SSP126 simulations (2100–2299), and extended SSP585 simulations (2100–2299). Although decadal-scale variability persists across all simulations, its amplitude generally decreases as the climate state becomes warmer, with the most pronounced reduction occurring during the extended SSP585 period (figure S2). To further examine the spatial structure of this weakening, we calculate the standard deviation of the overturning streamfunction at each latitude–depth grid point (figure S3). In piControl simulations, pronounced AMOC-DV is primarily concentrated in the subpolar North Atlantic, with the maximum variability of the multi-model ensemble mean

(MMEM) centered around 40°–50° N and extending from the upper ocean into the intermediate and deep ocean (figure 1(a)). The hatching denotes regions where significant 10–100 year variability is detected in at least seven ensemble members based on spectral analysis. As the climate state becomes warmer, AMOC-DV weakens across much of the subpolar North Atlantic, with the largest reductions occurring in regions of strong variability in the piControl climate (figure 1(b)). The hatching in figure 1(b) indicates areas where the changes in variability between climate states are statistically significant, demonstrating that the weakening is broadly consistent across the ensemble.

We next quantify the changes in AMOC-DV amplitude by calculating the standard deviation of the AMOC-DV index for each model and climate state (figure 1(c)). Despite large inter-model differences in the absolute magnitude of AMOC-DV under piControl, most models exhibit a decline in variability amplitude under future warming scenarios. Relative to piControl, the MMEM AMOC-DV amplitude decreases by 22.0% under extended SSP126 and 69.7% under extended SSP585, indicating a pronounced dependence of AMOC-DV amplitude on the magnitude of external forcing. Only one model shows an increase in AMOC-DV under extended SSP126, likely reflecting the influence of internal variability under relatively weak forcing. Overall, these results indicate a consistent climate-state dependence of AMOC-DV across CMIP6 models, with decadal overturning variability generally becoming weaker as anthropogenic warming intensifies.

In contrast to the amplitude changes, variations in the dominant timescale of AMOC-DV are less consistent across models, suggesting that its characteristic period is less sensitive to climate change than its amplitude (figure S4). The robust reduction in AMOC-DV amplitude across models and climate states motivates us to investigate the physical factors underlying this climate-state dependence.

3.2. Mean density flux as a constraint on AMOC-DV amplitude

The systematic weakening of AMOC-DV across climate states suggests that changes in the background buoyancy forcing may influence the energy available for sustaining overturning fluctuations. Surface density flux provides a physically motivated measure of this forcing, as surface heat and freshwater exchanges directly modify ocean density and influence the generation of density anomalies associated with deep convection (Marshall and Speer 2012, Tailleux 2013). We therefore examine whether variations in mean surface density flux can account for the climate-state dependence of AMOC-DV amplitude and provide a common constraint across CMIP6 models and climate states.

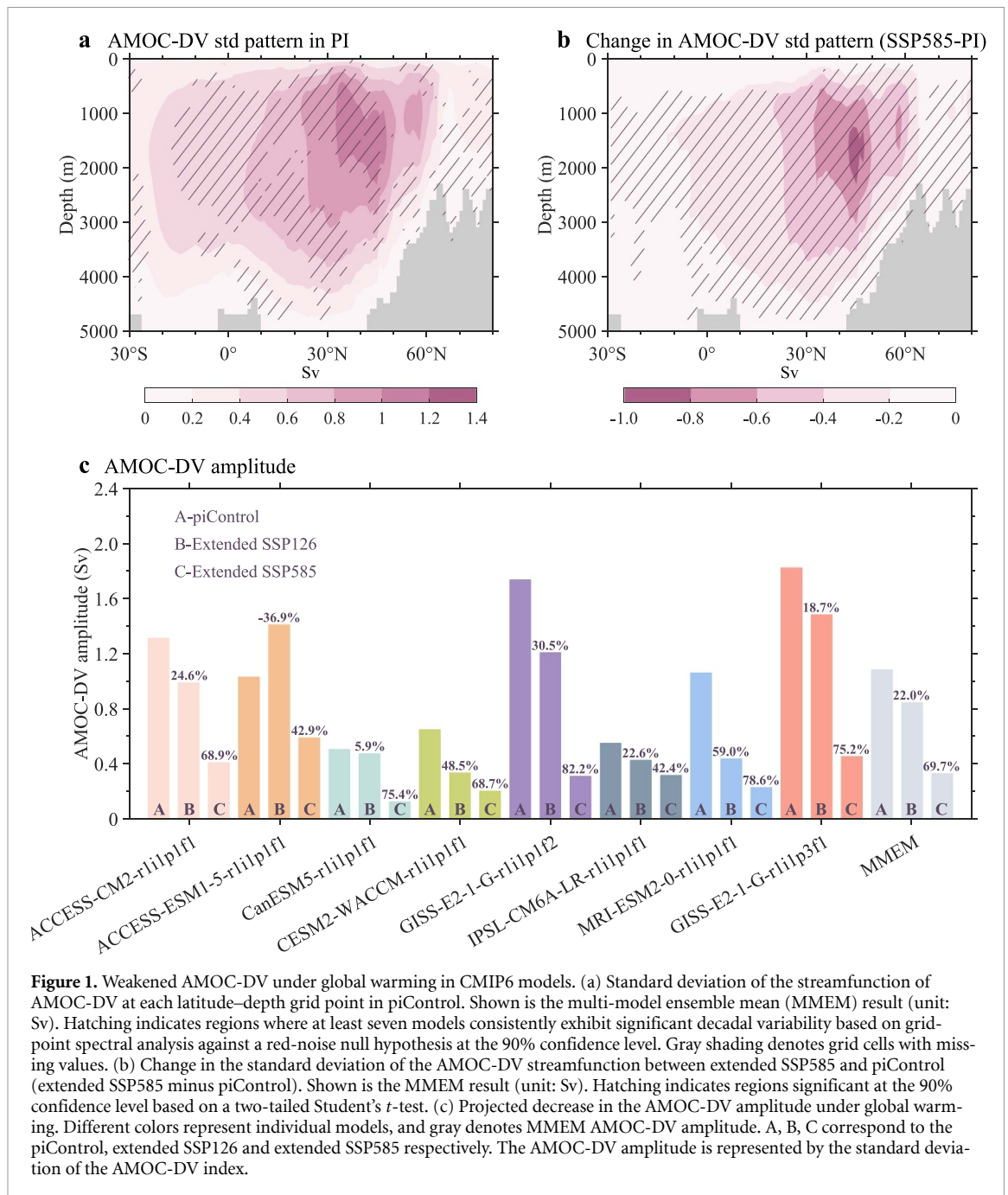
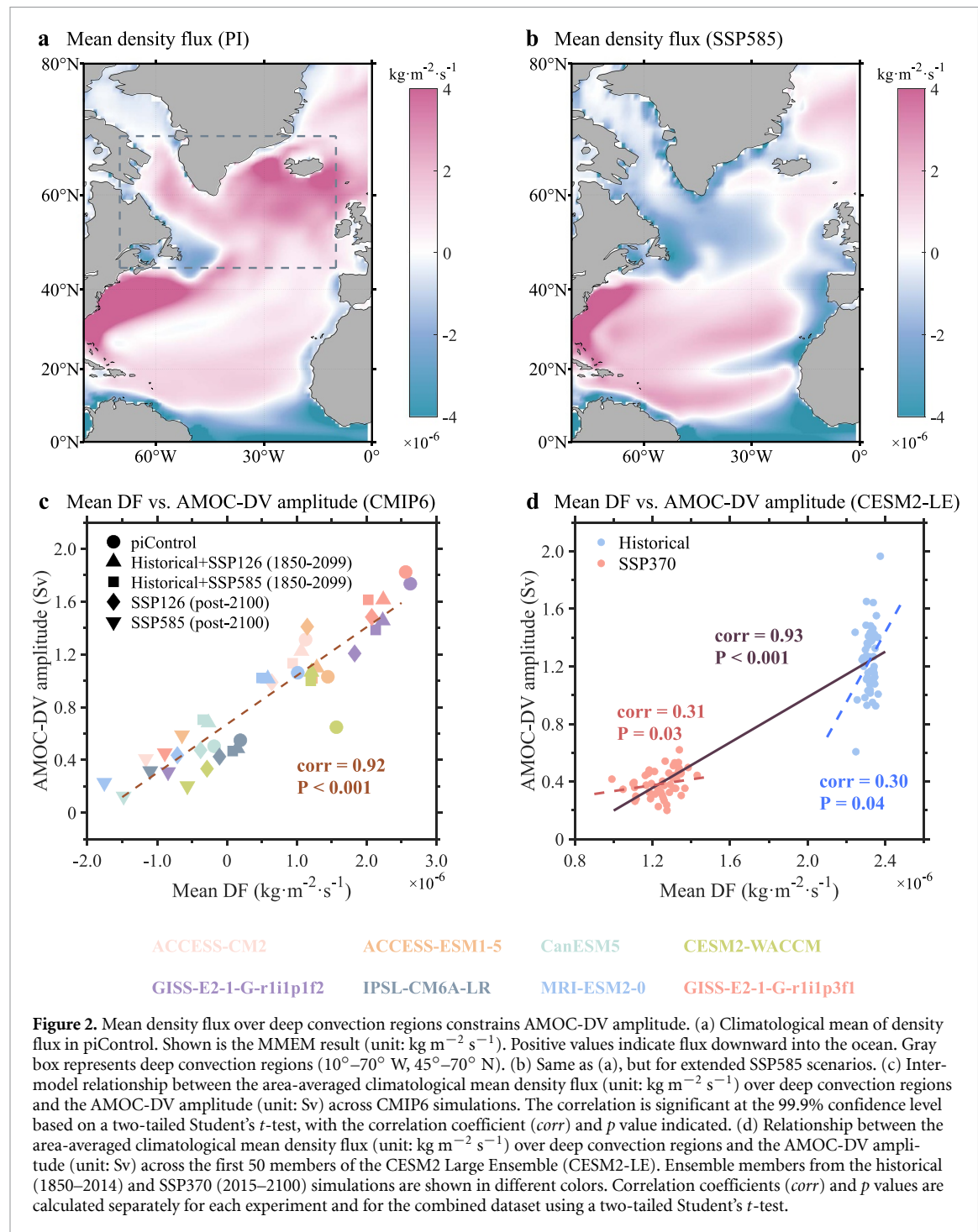


Figure 1. Weakened AMOC-DV under global warming in CMIP6 models. (a) Standard deviation of the streamfunction of AMOC-DV at each latitude–depth grid point in piControl. Shown is the multi-model ensemble mean (MMEM) result (unit: Sv). Hatching indicates regions where at least seven models consistently exhibit significant decadal variability based on grid-point spectral analysis against a red-noise null hypothesis at the 90% confidence level. Gray shading denotes grid cells with missing values. (b) Change in the standard deviation of the AMOC-DV streamfunction between extended SSP585 and piControl (extended SSP585 minus piControl). Shown is the MMEM result (unit: Sv). Hatching indicates regions significant at the 90% confidence level based on a two-tailed Student's *t*-test. (c) Projected decrease in the AMOC-DV amplitude under global warming. Different colors represent individual models, and gray denotes MMEM AMOC-DV amplitude. A, B, C correspond to the piControl, extended SSP126 and extended SSP585 respectively. The AMOC-DV amplitude is represented by the standard deviation of the AMOC-DV index.

Because AMOC-DV is closely linked to deep-water formation and convection in the subpolar North Atlantic, we focus on the regions where deep convection is most active to characterize the background density forcing relevant to AMOC-DV. We first examined the climatological distribution and future changes in MLD. In piControl simulations, the deepest mixed layers are located north of 45° N, particularly in the Labrador Sea, Irminger Sea, and Greenland-Iceland-Norwegian Seas, corresponding to regions of active deep convection (figures S5(a) and (c)). These regions also experience the most pronounced shoaling of the mixed layer under global warming (figures S5(b) and (d)). We therefore define 10°–70° W, 45°–70° N as the primary

deep-convection region (gray box in figure 2(a)), and calculate the mean surface density flux averaged over this region for subsequent analysis.

The spatial distribution of mean surface density flux reveals the changes in the background buoyancy forcing in a warming climate (figures 2(a) and (b)). In piControl simulations, positive density flux dominates much of the North Atlantic, particularly in mid-to-high latitudes, reflecting strong surface heat loss that increases surface water density and supports deep convection. In contrast, negative density flux is mainly confined to the tropical Atlantic, where net heat gain and freshwater input reduce surface density (figures S6(a) and (b)). Under global warming, the mean density flux over the North



Atlantic deep convection region decreases substantially (figure 2(b)). This reduction results from concurrent weakening of both thermal and haline contributions. The thermal component becomes less positive as enhanced downward longwave radiation and reduced sensible heat loss suppress oceanic surface cooling (figure S6(c)). Meanwhile, the haline contribution weakens over key convection regions, especially near continental margins, where increased ice melt and riverine freshwater input promote surface freshening (figure S6(d)). Together, these changes act to reduce the net density flux input into the ocean, weakening the background buoyancy forcing.

To examine whether changes in this background forcing are related to AMOC-DV amplitude across climate states, we analyze multiple CMIP6 simulation periods, including piControl simulations, concatenated historical and transient SSP simulations (1850–2099), and extended SSP simulations (2100–2299). We then assess the inter-model relationship between AMOC-DV amplitude and mean surface density flux for each period. This framework allows us to evaluate whether their relationship is maintained across transient warming and warmer quasi-equilibrium climate states. Across CMIP6 simulations spanning a broad range of climate states, AMOC-DV amplitude

exhibits a robust positive relationship with mean density flux ($r = 0.92$, $p < 0.001$; figure 2(c)). Models with stronger background buoyancy forcing generally exhibit larger AMOC-DV variability. Importantly, simulations from different climate states tend to converge toward a common scaling relationship, suggesting that the dependence of AMOC-DV amplitude on mean density flux is not confined to a particular climate state.

We further examined whether this relationship is also evident within individual models (figure S7). The within-model correlations are consistently positive, ranging from 0.79 to 0.99 across the eight experiments, with six reaching statistical significance at the 95% confidence level. The lack of statistical significance in the other two experiments may partly reflect internal variability, which can obscure the underlying relationship when the number of independent samples is limited. Nevertheless, the consistently positive within-model relationships provide complementary evidence that the association between mean density flux and AMOC-DV amplitude is not solely a consequence of systematic differences among models. Taken together, the inter-model and within-model results provide complementary perspectives on the mean–variance relationship. Across models under comparable climate conditions, differences in AMOC-DV amplitude are closely associated with variations in mean density flux, indicating that the background forcing state contributes to the inter-model spread in simulated variability. Within individual models, meanwhile, the progressive reduction in mean density flux from piControl to warmer climate states is generally accompanied by a reduction in AMOC-DV amplitude, suggesting that the same relationship also characterizes the forced evolution across climate states. We further test the sensitivity of this relationship to the choice of bandpass filter using a narrower 10–25 year band. The positive inter-model relationship between mean surface density flux and AMOC-DV amplitude remains strong ($r = 0.87$, $p < 0.001$; figure S8), indicating that the relationship is relatively insensitive to the choice of decadal bandpass range. This additional sensitivity test therefore provides further support for the robustness of the mean density flux constraint across models and climate states.

To further assess whether the mean–variance relationship extends beyond inter-model differences, we turn to the CESM2-LE. The large ensemble provides multiple realizations from the same climate model under identical external forcing, offering a complementary perspective on the roles of internal variability and externally forced changes in the background climate state. Specifically, it allows us to examine whether changes in AMOC-DV amplitude associated with different climate states can be distinguished from the spread generated by internal variability within a given climate state. Using the

first 50 ensemble members following the CMIP6 historical and SSP370 forcing pathways, we calculate mean surface density flux and AMOC-DV amplitude during the historical (1850–2014) and future warming (2015–2100) periods using the same approach as applied to the CMIP6 simulations. Within each period, differences among ensemble members in AMOC-DV amplitude are weakly related to variations in mean density flux, although the correlations remain statistically significant (historical: $r = 0.30$, $p = 0.04$; SSP370: $r = 0.31$, $p = 0.03$; figure 2(d)). This indicates that internal variability can introduce spread in AMOC-DV amplitude even under a similar externally forced climate state. Importantly, however, the ensemble exhibits a coherent shift from the historical to the SSP370 period, toward both lower mean surface density flux and lower AMOC-DV amplitude (figure 2(d)). When the two periods are considered together, they follow a strong scaling relationship ($r = 0.93$, $p < 0.001$).

The CESM2-LE analysis therefore provides a complementary test of the relationship identified in the CMIP6 inter-model comparison. In the CMIP6 analysis, differences in mean density flux and AMOC-DV amplitude are evaluated across models, whereas the CESM2-LE allows us to examine how this relationship evolves within a single model under a common set of external forcings. Within this framework, internal variability contributes to the spread of AMOC-DV amplitude around a given climate state, while the externally forced transition toward a warmer climate is accompanied by a systematic reduction in both mean density flux and AMOC-DV amplitude. This contrast suggests that the long-term weakening of AMOC-DV is more closely associated with the forced change in the background density-flux state than with internal variability alone. However, this interpretation is based on the CESM2-LE experiment alone and may not necessarily apply across different models or forcing scenarios. Because the analysis compares only the historical and SSP370 periods within CESM2-LE, it does not establish whether the same relationship holds under alternative forcing scenarios or in other large-ensemble experiments. Assessing the broader applicability of this relationship across models and ensemble configurations will require further investigation.

3.3. Conceptual interpretation of the mean–variance relationship

The robust scaling relationship between AMOC-DV amplitude and mean surface density flux indicates that the strength of decadal overturning variability is closely linked to the background buoyancy forcings. This relationship implies that the mean state may regulate how efficiently density flux anomalies are generated and converted into overturning anomalies. To interpret this relationship, we adopt a simplified conceptual framework in which AMOC-DV

is represented as a weakly damped and stochastically forced oscillator sustained by density flux anomalies in the subpolar North Atlantic (equation (6)). In deep convection regions, surface density flux anomalies generate buoyancy perturbations and can subsequently propagate into large-scale overturning adjustments. The evolution of AMOC-DV can therefore be idealized as a balance among surface buoyancy forcing, oceanic adjustment processes, and stochastic atmospheric forcing:

$$\frac{dM}{dt} = \alpha F' - \lambda M + \sigma \xi \quad (6)$$

where M denotes AMOC-DV, F' represents density flux anomalies in the subpolar gyre (SPG) region, α measures the sensitivity of overturning circulation to density flux decadal anomalies and can be interpreted as the response to unit density flux forcing, λ is an effective damping rate, and $\sigma \xi$ represents stochastic forcing associated with high-frequency atmospheric buoyancy flux variability. Under this framework, the equilibrium variance of AMOC anomalies can be approximated as (Text S1):

$$\text{Var}(M) \sim \frac{\alpha^2 \text{Var}(F') + \sigma^2}{2\lambda}. \quad (7)$$

This expression indicates that AMOC-DV amplitude is primarily controlled by four factors, including the sensitivity of overturning circulation to density flux forcing α , the variance of decadal density flux anomalies $\text{Var}(F')$, the effective damping rate λ and the contribution from stochastic atmospheric forcing σ . If the mean surface density flux in the SPG modulates any of these factors, it may provide a physical constraint on the equilibrium amplitude of AMOC-DV. We therefore examine whether the inter-model spread in AMOC-DV can be explained by differences in the mean buoyancy forcing state.

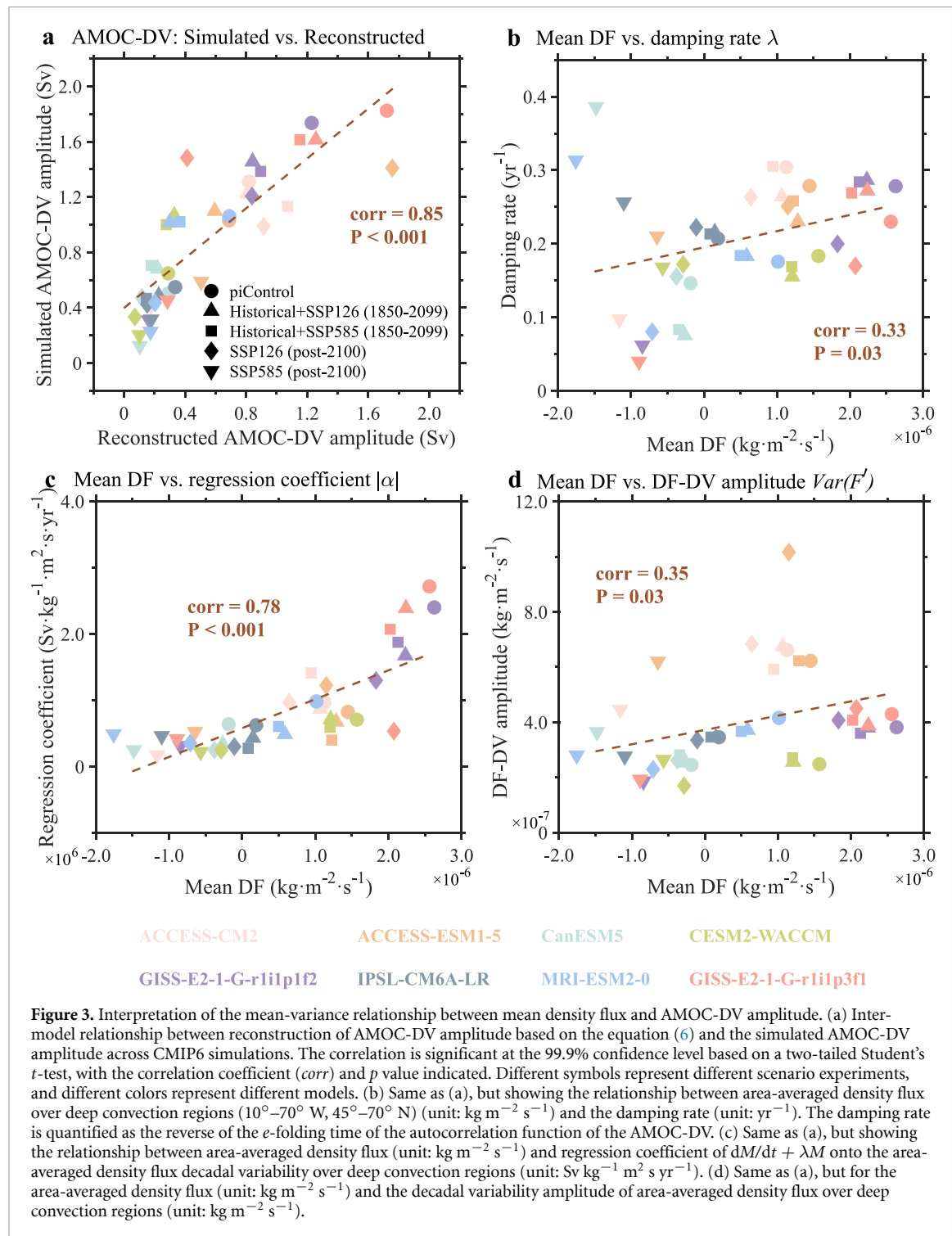
Before applying this framework to interpret the inter-model spread in AMOC-DV, we first evaluate whether the equation (7) captures the key characteristics of AMOC-DV amplitude across CMIP6 models. For each model and climate experiment, we calculate the AMOC-DV amplitude directly from the simulated AMOC time series. Meanwhile, we independently estimate the parameters in equation (7) from model outputs to reconstruct the variance of AMOC-DV. The reconstructed amplitudes closely match the directly simulated values, yielding a significant inter-model correlation of 0.85 across CMIP6 models and climate states (figure 3(a)). This agreement suggests that the simplified framework captures the dominant controls on AMOC-DV variance and provides a physically interpretable basis for understanding inter-model differences. Despite its simplicity relative to the fully coupled nonlinear ocean-atmosphere system, the framework retains the essential roles of surface density forcing, oceanic damping,

and stochastic atmospheric variability in regulating AMOC-DV amplitude.

To identify the physical pathways underlying this constraint, we next examine how mean density flux modulates each component controlling AMOC-DV variance in equation (7). We first assess whether the background mean density flux influences the damping rate λ , estimated as the inverse of the e -folding timescale of AMOC-DV autocorrelation. Across models, the changes in AMOC-DV autocorrelation and damping timescales from piControl to warming experiments show no consistent pattern (figure S9). Although λ exhibits a weak positive relationship with mean density flux across simulations ($r = 0.33$, $p = 0.03$; figure 3(b)), the changes in damping rate between piControl and warming scenarios are highly model-dependent, with some models exhibiting enhanced damping and others reduced damping. This lack of a consistent cross-model response suggests that variations in mean density flux do not systematically regulate the intrinsic adjustment timescale of AMOC-DV.

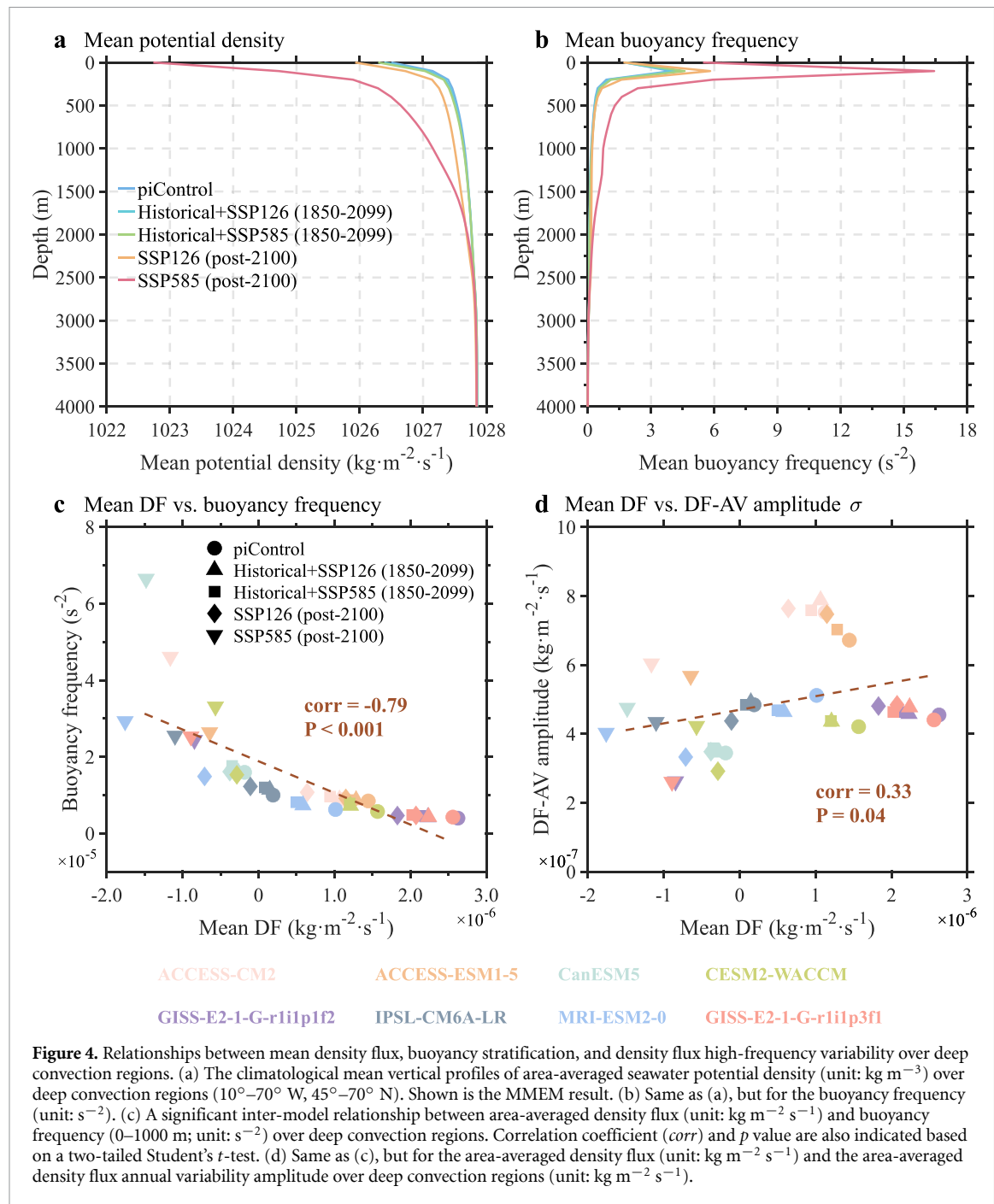
We next assess whether mean density flux modulates the regression coefficient α , which represents the sensitivity of AMOC-DV to surface density flux perturbations in the conceptual model. For each model, we estimate α by regressing the density-flux-driven component of AMOC tendency ($dM/dt + \lambda M$) against SPG density flux anomalies. The resulting regression slope provides an empirical measure of the overturning response to density flux forcing. Across CMIP6 models and climate states, α increases systematically with mean SPG density flux, exhibiting a significant positive correlation ($r = 0.78$, $p < 0.001$; figure 3(c)). Models with larger mean density flux, and thus stronger background surface densification, show stronger overturning response to density flux anomalies. This enhanced sensitivity is physically consistent with changes in the upper-ocean stratification and convection state of the SPG region (figures 4(a) and (b)). Stronger surface densification reduces upper-ocean stratification and facilitates the penetration of surface-generated density anomalies into the deep convection region, thereby strengthening their influence on the large-scale overturning circulation. The resulting increase in α indicates that a denser and more weakly stratified background state enhances the coupling efficiency between surface density forcing and AMOC variability. These results suggest that mean density flux contributes to the AMOC-DV amplitude scaling by increasing the dynamical sensitivity of the overturning system, rather than simply increasing the magnitude of external forcing.

We then examine whether differences in the variability of buoyancy forcing itself (F') contribute to the AMOC-DV scaling. The amplitude of decadal density flux variability exhibits a positive correlation with mean density flux across models and climate



states ($r = 0.35$, $p = 0.03$; figure 3(d)). Although this relationship is weaker than that found for α , the positive tendency suggests that regions with stronger background buoyancy forcing tend to sustain larger decadal density flux variability in deep convection regions. This relationship may arise because stronger mean surface densification is generally associated with more active deep convection and stronger air–sea coupling, which can enhance the sensitivity of surface buoyancy fluxes to atmospheric variability (figure 4(c)). Under such conditions, atmospheric variability is more effectively translated into

persistent density flux anomalies, contributing to larger decadal variability. Conversely, regions with weaker mean density flux exhibit weaker convection and reduced atmosphere–ocean coupling, which may limit the development and persistence of density flux anomalies. Furthermore, high-frequency stochastic density flux variability shows a weak positive relationship with AMOC-DV amplitude ($r = 0.33$, $p = 0.04$; figure 4(d)), suggesting that stronger stochastic buoyancy forcing may slightly enhance AMOC-DV but does not dominate the inter-model scaling.



Together, these results indicate that mean density flux constrains AMOC-DV amplitude primarily by modulating the effective gain of the overturning system. Specifically, stronger mean density flux enhances the sensitivity of AMOC-DV to density forcing α and, to a lesser extent, increases the amplitude of decadal density flux variability $\text{Var}(F')$ and high-frequency stochastic density flux variability, while exerting limited influence on the damping rate. In this perspective, the mean state determines how efficiently surface buoyancy perturbations are transmitted into large-scale overturning anomalies. As a result, models with stronger climatological density flux operate in a higher-gain regime and exhibit systematically larger AMOC-DV

amplitudes across both preindustrial and warming climates.

4. Discussions and conclusions

Our analysis reveals that, across CMIP6 models and climate states, the amplitude of AMOC-DV is closely constrained by the mean surface density flux over the subpolar North Atlantic. We demonstrate that this constraint operates through two synergistic pathways. First, the mean buoyancy forcing regulates the dynamical gain of the overturning circulation. By setting the mean AMOC strength and upper-ocean stratification, it controls how efficiently density flux

anomalies are converted into circulation anomalies. In a more strongly forced, dynamically ‘tense’ system, a given density flux perturbation can induce a larger AMOC variability response. Second, the mean buoyancy state modulates the variance of the buoyancy forcing itself. Regions with stronger mean density flux tend to host more active air-sea interaction, thereby producing larger-amplitude density flux fluctuations that drive the system. Together, these two pathways establish a mean–variability coupling, the climatological buoyancy forcing simultaneously controls both the efficiency of the oceanic response and the amplitude of the density flux decadal fluctuations. As a result, models with stronger mean density flux operate in a higher-gain, more strongly forced regime, and therefore exhibit systematically larger AMOC-DV amplitude.

Under global warming, reduced surface heat loss and enhanced freshwater input weaken the mean density flux in deep convection regions. This shifts the system toward a lower-gain and weaker forced regime, suppressing the amplitude of internally generated decadal variability. This energetic perspective provides a physically grounded explanation for the projected decline of AMOC-DV under warming. More broadly, our results suggest that accurately representing mean buoyancy forcing in the subpolar North Atlantic is essential not only for simulating the mean overturning circulation, but also for constraining the magnitude and predictability of its variability across climate states.

These findings have several implications. First, they suggest that variations in AMOC-DV across past and future climates may be interpretable through changes in background buoyancy forcing, offering a unifying mechanism across climate states. Second, because surface density flux can be estimated from surface heat and freshwater flux observations or reanalysis data, it may provide a useful diagnostic for evaluating simulated AMOC variability strength. Finally, the projected reduction in AMOC-DV implies weakened multidecadal modulation of Atlantic climate variability, with potential consequences for SST patterns, hydroclimate variability, and related climate extremes under future warming.

Despite the robustness of the identified relationship, several limitations remain. First, the proposed constraint is inferred from an emergent relationship across CMIP6 models and climate states rather than from controlled perturbation experiments. Although the consistency of the relationship across different forcing scenarios supports a physical interpretation, statistical relationships alone cannot establish causality uniquely. Targeted sensitivity experiments that independently perturb surface buoyancy forcing would provide a more direct test of the proposed mechanism. Second, our analysis is based on CMIP6 models, in which key

processes governing deep convection and overturning variability, including vertical mixing, overflow transport, and mesoscale eddy effects, remain parameterized to varying degrees. Such structural differences may influence both the simulated mean density flux and AMOC response. Evaluating whether the identified constraint persists in higher-resolution and eddy-rich simulations will therefore be an important direction for future work. Finally, direct observational validation remains challenging because existing AMOC observations are too short to robustly characterize multidecadal variability. Whether the emergent mean–variability relationship identified across climate models also applies to the real climate system will require longer observational records and improved constraints on historical AMOC variability.

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Data availability statement

CMIP6 data used in this study can be downloaded from the website below: <https://esgf-node.llnl.gov/projects/cmip6/>.

CESM2 Large Ensemble (CESM2-LE) data can be downloaded from the website below: <https://gdex.ucar.edu/datasets/d651056/>.

Supplementary data 1 available at: <http://doi.org/10.1088/1748-9326/aea7b8/data1>.

Code availability

The code for climate model data processing and analyses is available via Zenodo at: <https://zenodo.org/records/21497465>.

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